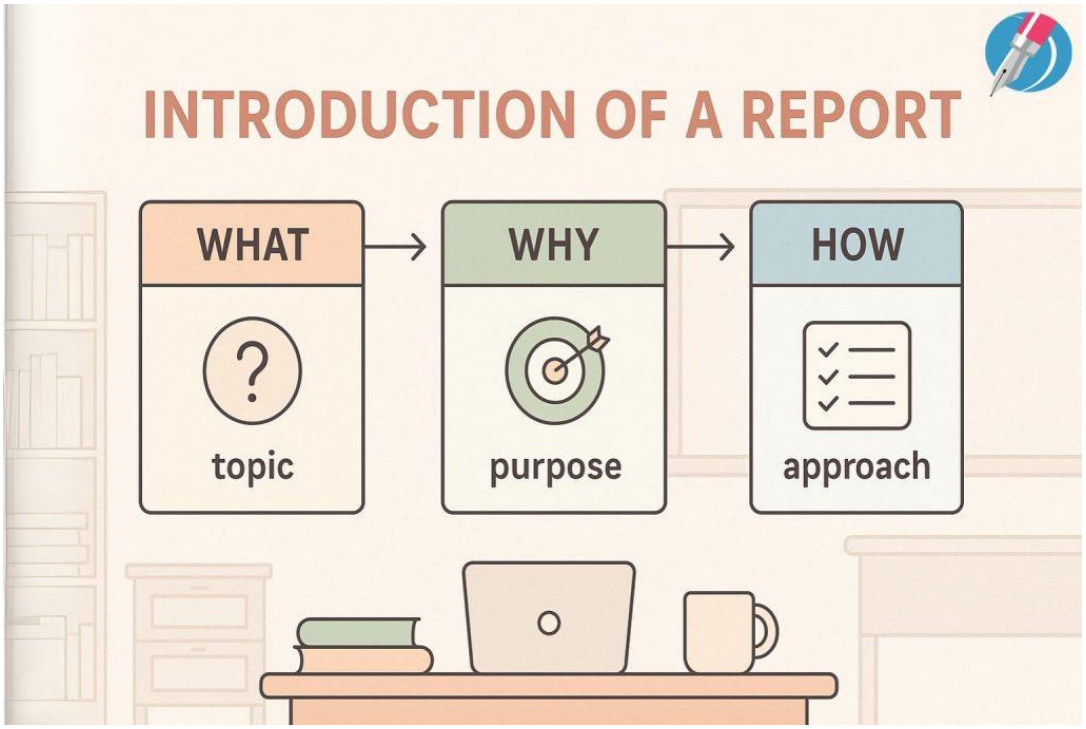
1. Introduction (What? Why? How?)

The introduction should explain what the project is about, why it is done, and how it is going to be carried out. It sets the scene for the reader and helps them understand the problem and the goal of your technical solution.



Structure (the what):

1. **Background / Context:**
    - Explain the situation or industrial context (e.g., workshop, process, client requirement).
    - Briefly describe the problem or need that motivated the project.
2. **Objectives (the why):**
    - State clearly the main goal of your project: mention what you want to design, build, or improve.

How to write objectives:

Start with a general sentence that introduces the objectives. Then make a list with the objectives of your project, from most important to least important.

The objectives usually follow the SMART framework, that is, they must be:

- Specific, Measurable, Achievable, Relevant, Time-bound (if applicable)

3. **Scope and approach (the how):**
    - Summarize the main steps or methods you are going to use (e.g., design, material selection, manufacturing, testing).
    - Mention tools, technologies, or software (e.g., CAD, CNC, simulation, etc.) if relevant.
4. **Expected outcome:**
    - Say briefly what result or benefit is expected (e.g., higher efficiency, safer system, better precision, etc.).

Useful phrases:

- *This project aims to…*
- *The main objective is to design and build…*
- *The project was developed in response to…*
- *The work was carried out in the mechanical workshop using…*
- *The expected result was…*

3. Conclusion (What was achieved? What was learned?)

The conclusion summarizes the main results, difficulties, and learning outcomes. It shows reflection and professional awareness.

Structure:

1. **Summary of results:**
    - Briefly describe what was achieved.
    - Mention whether the objectives were met.
2. **Evaluation:**
    - Reflection and self-evaluation of the project
    - Evaluate how the group has worked as a team.
    - Mention problems faced or improvements made.
3. **Learning outcomes and future work:**
    - Explain what you learned technically and professionally.
    - Suggest possible improvements or future developments.

Useful phrases:

- *The main objective of the project was achieved by...*
- *During the process, we faced challenges related to...*
- *The final result met the required specifications...*
- *We learned to apply design and manufacturing techniques such as...*
- *In future projects, it would be interesting to improve...*

1. Introduction

1.1. Background / Context

In healthcare, monitoring patients' vital signs is very important to understand their health condition and detect possible problems early. Vital signs such as heart rate, body temperature, and blood oxygen level provide useful information about how the human body is functioning.

(This is an example of the situation or industrial context)

In many hospitals and home-care situations, these measurements are still taken manually or only from time to time. This can lead to delayed detection of health problems and reduced quality of patient care.

(This is the problem that motivates the project)

This project is developed to answer the need for a simple and reliable biomedical monitoring system that can continuously measure basic vital signs. The project is carried out in a biomedical engineering context, where accuracy, safety, and reliability are essential. The work is performed in the biomedical engineering laboratory using standard biomedical sensors and electronic components.

(We say what we are going to do to solve the problem)

1.2. Objectives

The main objective of this project is to design and develop a biomedical system for real-time monitoring of vital signs.

The specific objectives of the project are:

- To study the main physiological parameters used in basic patient monitoring.
- To select suitable sensors for measuring heart rate, body temperature, and oxygen saturation.
- To design and assemble an electronic system for data acquisition.
- To develop software to process and display the measured data in real time.
- To test and validate the system under simulated conditions.

1.3. Scope and approach

The project starts with the study of biomedical requirements and basic safety rules related to patient monitoring. Suitable sensors are then selected based on accuracy and compatibility with the electronic system.

The hardware system is designed using a microcontroller and signal conditioning circuits. Software is developed to read sensor data, process the signals, and display the results.

Finally, the complete system is tested and compared with reference measurements to check correct operation and reliability.

1.4. Expected outcome

The expected result of this project is a functional biomedical monitoring system capable of measuring and displaying vital signs in real time. The system is expected to improve patient monitoring by providing continuous and reliable data while respecting basic biomedical safety requirements.

2. Conclusion

2.1. Summary of results

The main objective of the project was to design and develop a biomedical system for real-time monitoring of vital signs. This objective was successfully achieved, as the final system is capable of measuring heart rate, body temperature, and oxygen saturation and displaying the data in real time.

All the specific objectives defined at the beginning of the project were fulfilled. The study of physiological parameters was completed, allowing the identification of the most relevant vital signs for basic patient monitoring. Suitable sensors were successfully selected for each parameter, ensuring compatibility with the electronic system.

The design and assembly of the electronic data acquisition system were achieved, enabling correct signal capture and processing. In addition, the software was developed to process and display the data in real time, meeting the corresponding objective. Finally, the system was tested under simulated conditions, confirming its correct operation and validating the overall design.

Therefore, both the general and specific objectives of the project were accomplished, resulting in a functional and reliable biomedical monitoring system.

2.2. Evaluation

From a technical point of view, the system performed satisfactorily, providing continuous and relatively accurate measurements of the selected vital signs. Although some issues such as signal noise and calibration errors were encountered during the development process, these were progressively reduced through software adjustments and improved sensor placement.

From a project perspective, the group worked effectively as a team by dividing tasks according to each member's strengths, such as hardware design, programming, and testing. Good communication and collaboration allowed the team to solve problems efficiently and keep the project on schedule.

However, some difficulties were experienced, particularly in integrating the different components and ensuring stable signal acquisition. These challenges highlighted the importance of careful planning and iterative testing. As a result, several improvements were made during the project, leading to a more reliable final system.

Overall, the objectives were achieved, and the project demonstrated both technical functionality and effective teamwork, although there is still room for further optimization and refinement.

2.3. Learning outcomes and future work

During the development of this project, several important technical and professional skills were acquired. From a technical perspective, the project allowed the application of knowledge related to biomedical sensors, electronic circuit design, signal processing, and microcontroller programming. It also provided practical experience in integrating hardware and software into a complete functional system.

From a professional point of view, the project helped to develop teamwork, problem-solving, and project management skills. Working in a group required clear communication, task organization, and collaboration to overcome technical challenges and meet project deadlines. In addition, the importance of accuracy, safety, and reliability in biomedical applications became evident throughout the project.

For future work, several improvements could be considered to enhance the system. For example, adding wireless communication would allow remote monitoring of patients. Data storage functionality could also be implemented to keep a record of measurements over time. Furthermore, developing a more user-friendly interface would improve usability in real healthcare environments.

Overall, these potential developments would increase the functionality and applicability of the system, making it more suitable for practical biomedical use.